

ARINDAM SINGHA

Robotics AI Engineer · Autonomous Systems · ROS2 · Multi-Agent Control · Deep Learning

Birbhum, West Bengal, India · singha.drarindam@gmail.com · +91-8967859942
linkedin.com/in/arindam-singha · researchgate.net/profile/Arindam-Singha

PROFESSIONAL PROFILE

Ph.D.-qualified Robotics AI Engineer with 8+ years of experience architecting **intelligent control systems**, **multi-agent robotic frameworks**, and **production-grade ROS2 deployments** for UAV and UGV platforms. Proven expertise in sensor fusion, motion planning (MoveIt2, NAV2), and real-time perception pipelines. Delivered a microservice-based vision platform to production and led the conversion to a ROS2-based architecture for industrial automation. Seeking to drive autonomous system development in a European robotics or automotive R&D environment.

CORE COMPETENCIES

Robotics & Autonomy:	ROS2, Gazebo, Rviz, URDF, TF2, MoveIt2, NAV2, DDS, LiDAR, Sensor Fusion, SLAM, Path Planning
Control Systems:	Adaptive RBFNN Control, Lyapunov-Based Control, Leader-Follower Formation, Trajectory Tracking, MPC
AI & Perception:	Deep Learning, Computer Vision, YOLOv8, Semantic Segmentation, Object Detection, Keypoint Detection
Deployment & MLOps:	Docker, Azure, Triton Inference Server, Webots, Digital Twin, CI/CD, Microservices, gRPC
Programming:	Python, C, MATLAB, SCILAB
Hardware:	Raspberry Pi, Arduino, Pixhawk Flight Controller, Embedded Systems

PROFESSIONAL EXPERIENCE

Lead Engineer — AI & Robotics

Jul 2024 – Present

Magna Steyr India, Pune, India

- Architected and deployed a **ROS2-based vision platform** to production for automated defect detection, replacing a legacy microservice architecture and reducing system latency by 40%.
- Developed a **three-wheel differential drive vehicle framework** using ROS2 and NAV2 for autonomous indoor navigation and obstacle avoidance.
- Designed and deployed **deep learning models** (YOLOv8, encoder-decoder) for multi-class defect detection and semantic segmentation on automotive components.

- Built **Python-based real-time data pipelines** for production monitoring, enabling data-driven decision-making on the factory floor.
- Deployed model ensembles using **Triton Inference Server** for high-throughput, low-latency inference in industrial environments.
- Led the development of an **agentic-RAG chatbot** (Langchain, LangGraph) to auto-generate software test cases from engineering requirements, reducing manual effort by 60%.

Senior Engineer — Perception & Simulation

Feb 2022 – Jun 2024

Magna Steyr India, Pune, India

- Developed a **digital twin of the Mobileye Q3 camera sensor** in the Webots simulation platform for virtual ADAS validation.
- Engineered a **ROS2-Webots bridge** enabling seamless hardware-in-the-loop (HIL) testing and sensor data streaming.
- Built **microservices with gRPC** for inter-process communication between perception, planning, and control modules.
- Managed **Azure cloud deployments** with Docker containerization, ensuring 99.9% uptime for vision-based inspection systems.

Senior Research Fellow

Jan 2018 – Dec 2021

National Institute of Technology Sikkim, India

- Developed an **adaptive RBFNN-driven control framework** for multi-UAV formation control under external disturbances and directed/undirected network topologies.
- Designed **leader-follower formation controllers** with Lyapunov stability guarantees for quadrotor UAV swarms.
- Published **6+ peer-reviewed papers** on neural network-based robotic control and multi-agent coordination.

Junior Research Fellow

Jan 2016 – Dec 2017

National Institute of Technology Sikkim, India

- Developed **neural network-based path planning algorithms** for single and multiple UGVs in grid-based environments.
- Built a **trajectory tracking framework** for multiple UAVs using Kalman filtering and nonlinear control techniques.

EDUCATION

Ph.D., Intelligent Networked Robotic Systems

2016 – 2021

National Institute of Technology Sikkim, India

Dissertation: Adaptive Neural Network-Based Control for Multi-UAV Formation under Network Constraints and External Disturbances

M.E., Electronics, Instrumentation & Control

2012 – 2014

Thapar University, India

B.Tech, Applied Electronics and Instrumentation Engineering

2008 – 2012

Abacus Institute of Engineering and Management (MAKAUT), India

SELECTED PUBLICATIONS

A. Singha et al., "Adaptive Neural Network based Quadrotor UAV Formation Control under External Disturbances," *Aerospace Science and Technology* (Springer), 2021.

A. Singha et al., "Lyapunov based Trajectory Tracking Controller for a Quadrotor UAV with Nonholonomic Constraints," *e-Prime – Advances in Electrical Engineering, Electronics and Energy* (Springer), 2024.

A. Singha et al., "UGV-UAV Coordination: Take-off, Landing and Formation Control," *World Automation Congress* (IEEE), 2021.

A. Singha et al., "Neural Dynamics Based Complete Grid Coverage by Single and Multiple Mobile Robots," *SN Applied Sciences* (Springer), 2020.

Full list of 12 peer-reviewed publications available on ResearchGate.

KEY PROJECTS

Intelligent Networked Robotic Systems (Ph.D. Thesis) 2016 – 2021

Comprehensive framework for multi-UAV formation control, trajectory generation, and UGV-UAV coordination (take-off, landing, and cooperative tasks). Validated in Gazebo and on Pixhawk hardware.

Benchmarking Motion Planning Algorithms 2024 – 2025

Benchmarked RRT, RRT-Connect, and PRM for a 6DoF robot manipulator using ROS2 and MoveIt2 for pick-and-place operations.

AWARDS & GRANTS

- Visvesvaraya PhD Scholarship — Digital India Corporation, Ministry of Electronics & IT, Govt. of India
- MHRD GATE Scholarship — Ministry of Human Resource Development, Govt. of India
- Team of the Month — Magna Steyr India (Oct 2023, Apr 2025)
- Employee of the Month — Magna Steyr India (Dec 2022)

Mobility & Work Authorization

Open to positions in the Netherlands, Germany, and EU-wide. Eligible for EU Blue Card sponsorship (recognized Indian STEM PhD + qualifying employment contract). Available for immediate relocation.